

Case Study I: Inter-Terminal Transfer Optimization

Background

RSGT Bangladesh operates two yards: Patenga Container Terminal (PCT) and the South Container Yard (SCY), located 3.5 km apart. To facilitate operations, RSGT utilizes three transport vendors (Showkat Trading, Siam Enterprise, and SI Chowdhury) to shuttle loaded containers from PCT to SCY and return empty containers from SCY to PCT.

The shuttle cycle consists of six specific timestamped events:

PCT Yard: Loading process inside PCT.

PCT Out: Gate-out from PCT.

SCY In: Gate-in at SCY.

SCY Yard: Unloading/Loading process inside SCY.

SCY Out: Gate-out from SCY.

PCT In: Gate-in at PCT (Cycle complete).

Management Objectives

The Operations Manager requires a data-driven review to answer:

Vendor Performance: Which trucking company is the most efficient?

Temporal Analysis: How does the time of day (Morning, Evening, Night) affect travel time between terminals?

Bottleneck Identification: Which specific stage of the cycle accounts for the highest delays?

Technical Tasks

Task I: Data Engineering (Excel / Power Query)

Import the provided raw data using Power Query.

Automation: The solution must be built using the "From Folder" method. The model should automatically refresh and clean new data added to the source folder.

Data Cleaning: Handle null values, standardize timestamp formats, and handle logical errors (e.g., an "Out" time occurring before an "In" time).

Task 2: Dynamic Visualization (Power BI)

Develop a high-impact dashboard that includes:

KPI Tiles: Average Cycle Time, Total Moves per Vendor, and Peak Delay Stage.

Time-Slicing: Ability to filter by Shift (Morning/Evening/Night) or Date.

Comparative Analysis: A visual comparing the 3 vendors across different stages.

Innovation: Include at least one "Advanced Insight"

Task 3: Operational Analysis & Strategy

Based on your findings, provide a brief written proposal (300 words max) identifying the primary bottleneck.

Suggest two actionable improvements to reduce the overall cycle time.

Deliverables

Raw Input File: The Excel file used as the data source.

Business Intelligence File: The .pbix file containing the data model and dashboard.

Documentation: A PDF file detailing:

The logic behind your data cleaning steps.

Any assumptions made regarding missing timestamps or shift timings.

Your final operational recommendations.

Note:

If you feel any information is missing, make a logical assumption and mention it in your deliverables.